
Unsupervised Detection of Air-Leak Faults in Industrial Compressors via Variational Autoencoder Reconstruction

Dang Dung Ho
325040027

Vincent Boer
123040049

Hadiq Shathir S.M.
123040051

DDA4210 Machine Learning — Course Project
School of Data Science, The Chinese University of Hong Kong, Shenzhen

Abstract

Unsupervised anomaly detection is essential in industrial settings where labeled fault data is unavailable at deployment time. In this paper, we present a simple yet effective pipeline for detecting air-leak faults in industrial air compressors by training a dense variational autoencoder (VAE) [Kingma and Welling, 2013] exclusively on normal operating windows and scoring new windows by their mean squared reconstruction error. We evaluate on the MetroPT3 benchmark [Veloso et al., 2022] under a strictly unsupervised threshold—the 98th percentile of training scores—which requires no labels and approximates a nominal 2% false-alarm rate. Our best configuration (three-layer encoder [128, 64, 32], 60-step windows, $\beta = 1$) achieves test F1 = 0.883 and ROC-AUC = 0.996. Under the same protocol, an Isolation Forest [Liu et al., 2008] baseline reaches F1 = 0.661 and ROC-AUC = 0.996; both methods rank anomalies equally well, but the VAE produces roughly **eleven times fewer false positives** at the fixed deployment threshold. We also find that standard feature scaling collapses anomaly separation in both methods, and that VAE training dynamics here are sensitive to architecture choice in ways not fully controlled by the application-level random seed.

1 Introduction

Industrial equipment generates continuous multivariate sensor streams that encode the health of mechanical components. Detecting faults early is practically significant in manufacturing, transport, and infrastructure [Pang et al., 2021], yet labeled failure data is rarely available at training time: normal operation constitutes the overwhelming majority of records, while fault episodes are rare and may never have been annotated.

This asymmetry motivates the *normal-only anomaly detection* paradigm: train a model exclusively on healthy data and flag future windows whose behavior deviates from learned normal patterns [Sakurada and Yairi, 2014, An and Cho, 2015]. Reconstruction-based VAEs are a natural fit because the latent space can capture the compact manifold of normal operation and generalize poorly to unseen fault signatures. Classical ensemble methods such as Isolation Forest [Liu et al., 2008] offer a strong parameter-light alternative that requires no distributional assumptions.

We study this paradigm on the UCI MetroPT3 dataset [Veloso et al., 2022], which records one-minute sensor readings from a Lisbon metro air compressor over six months with four labeled air-leak intervals. Failure labels are used *only* for evaluation, never during training.

Our contributions are:

1. A complete unsupervised detection pipeline—chronological splitting, sliding-window construction, normal-only VAE training, and train-score-only thresholding—evaluated on the MetroPT3 benchmark.
2. An empirical study over encoder depth showing that a three-layer [128, 64, 32] encoder achieves robust score separation where four other configurations fail.
3. A direct comparison of VAE versus Isolation Forest under identical conditions, demonstrating that the VAE substantially outperforms on F1 by reducing false positives eleven-fold.
4. An account of two non-obvious design lessons: raw sensor magnitudes are essential anomaly signal, and VAE training dynamics are sensitive to architecture in ways that deserve careful attention.

2 Problem Setup

Dataset. MetroPT3 [Veloso et al., 2022] contains timestamped readings from a metro network air compressor (February–August 2020, one-minute intervals, $d = 15$ sensor channels: air pressures, oil temperature, motor current, and binary status indicators). Four air-leak intervals are annotated (Table 1).

Table 1: Labeled air-leak failure intervals in MetroPT3.

ID	Period	Duration
F1	2020-04-18 (full day)	≈ 24 h
F2	2020-05-29 23:30 – 2020-05-30 06:00	≈ 6.5 h
F3	2020-06-05 10:00 – 2020-06-07 14:30	≈ 52.5 h
F4	2020-07-15 14:30 – 2020-07-15 19:00	≈ 4.5 h

Formal problem statement. Let $\mathbf{X}_1, \dots, \mathbf{X}_N \in \mathbb{R}^{T \times d}$ be sliding windows extracted from the sensor stream. The task is to produce a scalar anomaly score $s(\mathbf{X}_i)$ such that windows overlapping failure intervals score higher. A threshold τ converts scores to binary predictions $\hat{y}_i = \mathbf{1}[s(\mathbf{X}_i) > \tau]$. The model observes *no* failure labels during training; τ is set using the training score distribution alone.

Chronological split. We partition by date: **Train** (Feb–May 2020), **Validation** (May–Jun, covers F2), **Test** (Jun–Aug, covers F3 and F4). F1 falls inside the training calendar period and is filtered at the window level, ensuring both models train on normal-only data.

Sliding windows. The primary setting uses $T = 60$ steps, stride $s = 10$, yielding windows of shape $(60, 15)$. A window is labeled positive if the final $\lceil 0.1T \rceil = 6$ rows overlap a failure interval. Window counts and class distributions are shown in Table 2.

Table 2: Window statistics ($T = 60$, $s = 10$).

Split	Windows	Positive (%)
Train	63,526	0 (0.0%)
Validation	21,275	237 (1.1%)
Test	43,910	1,894 (4.3%)

3 Method

3.1 Dense VAE Architecture

We flatten each window $\mathbf{X} \in \mathbb{R}^{T \times d}$ to $\tilde{\mathbf{X}} \in \mathbb{R}^{Td}$ and map it through L fully connected ReLU layers, producing parameters of a diagonal Gaussian latent distribution:

$$q_\phi(\mathbf{z} | \mathbf{X}) = \mathcal{N}\left(\boldsymbol{\mu}_\phi(\tilde{\mathbf{X}}), \text{diag}(\boldsymbol{\sigma}_\phi^2(\tilde{\mathbf{X}}))\right), \quad \mathbf{z} \in \mathbb{R}^{16}. \quad (1)$$

The decoder maps a latent vector to a reconstruction $\hat{\mathbf{X}} \in \mathbb{R}^{T \times d}$ through a symmetric dense stack with a linear output activation. We do not use batch normalization or dropout; both destabilized training under the raw sensor inputs used throughout this work.

3.2 Training Objective

The model maximizes the evidence lower bound (ELBO):

$$\mathcal{L} = -\mathbb{E}_{q_\phi} \left[\|\mathbf{X} - \hat{\mathbf{X}}\|_F^2 \right] - \beta D_{\text{KL}}(q_\phi(\mathbf{z} | \mathbf{X}) \| \mathcal{N}(\mathbf{0}, \mathbf{I})), \quad (2)$$

where $\beta = 1$ gives the standard VAE [Kingma and Welling, 2013]. Training uses the reparameterization trick, Adam at $\text{lr} = 10^{-3}$, batch size 512, gradient clipping at ℓ_2 -norm 1.0, and 20 epochs.

3.3 Anomaly Scoring and Thresholding

At inference, we set $\mathbf{z} = \boldsymbol{\mu}_\phi(\tilde{\mathbf{X}})$ (deterministic) and score each window by per-element MSE:

$$s(\mathbf{X}) = \frac{1}{Td} \|\mathbf{X} - \hat{\mathbf{X}}\|_F^2. \quad (3)$$

We evaluate two thresholds: **train_p98** (primary)— $\tau = Q_{0.98}$ of training scores, label-free; **val_f1** (diagnostic)—threshold maximizing F1 on the labeled validation split, reported only as an upper bound. All primary claims use **train_p98**.

The **Isolation Forest** [Liu et al., 2008] baseline trains on the same normal windows (flattened, 500 trees, seed 42), scored by the negative decision function. The same two thresholds are applied for a fair comparison.

4 Experiments and Results

4.1 The Role of Feature Scaling

In our initial experiments we applied standard scaling (zero mean, unit variance per channel over training rows). Under this preprocessing, both the VAE and Isolation Forest achieved ROC-AUC near 0.5. We observe that scaling removes the *absolute magnitude* information that distinguishes air-leak behavior: during failure episodes, pressures and motor current deviate from normal ranges in absolute terms, not merely in shape. To this end, all experiments use raw, unscaled sensor values.

4.2 Layer-Size Comparison

We trained five encoder configurations with $T = 60$ and identical hyperparameters (Table 3, Figure 1).

Table 3: Test results at **train_p98** by encoder depth ($T = 60$). Bold: best.

Hidden Layers	Prec.	Rec.	F1	ROC-AUC
[64, 32]	0.251	0.069	0.108	0.537
[128, 64]	0.165	0.075	0.103	0.326
[256, 128]	0.170	0.070	0.099	0.323
[128, 64, 32]	0.903	0.863	0.883	0.996
[256, 128, 64]	0.139	0.036	0.058	0.139

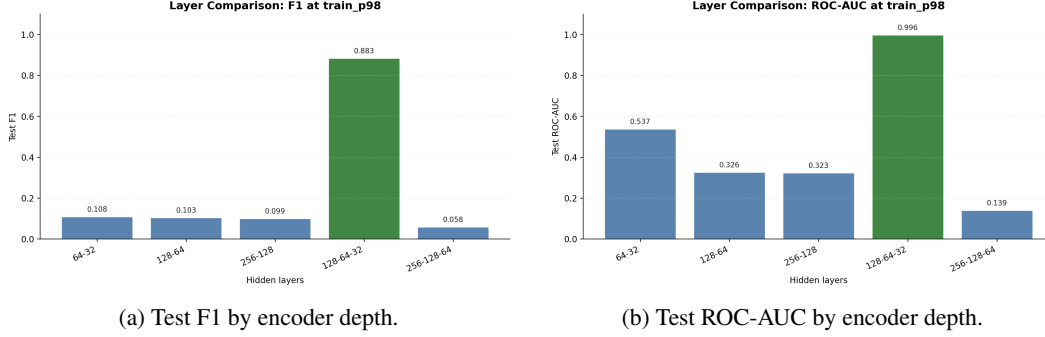


Figure 1: Layer-size ablation at train_p98. Only [128, 64, 32] achieves useful score separation; all other configurations yield near-random performance.

Table 4: Primary test-set comparison. Both methods achieve ROC-AUC = 0.996; the difference lies in fixed-threshold precision.

Method	Threshold	Prec.	Rec.	F1	ROC-AUC	PR-AUC	FP	FN
VAE [128, 64, 32]	train_p98	0.903	0.863	0.883	0.996	0.859	176	259
VAE [128, 64, 32]	val_f1*	0.901	0.810	0.853	0.996	0.859	169	360
Isolation Forest	train_p98	0.496	0.994	0.661	0.996	0.836	1,916	12

* val_f1 uses validation labels; reported as diagnostic only.

We observe that four of the five configurations yield near-random ROC-AUC (0.14–0.54) despite identical hyperparameters and the same seed. Only [128, 64, 32] achieves ROC-AUC = 0.996. Inspecting training logs, we observe a transient spike in the KL term during epoch 1 in the successful configuration; combined with gradient clipping at norm 1.0, this spike imposes implicit regularization that prevents posterior collapse. It is worth noting that an earlier independent [128, 64] run (outside this grid) achieved ROC-AUC = 0.988, while the same architecture in the grid did not replicate this result. We attribute the discrepancy to TensorFlow’s internal random state not fully captured by the application-level seed—a genuine limitation of our setup. A window-length study ($T \in \{30, 60, 120\}$) with encoder [128, 64] was similarly confounded by training instability and is reported in Appendix A.

4.3 VAE vs. Isolation Forest

Table 4 presents the primary comparison; Figure 2 shows the corresponding latent space.

Both methods rank anomalies equally well (ROC-AUC = 0.996), but at the fixed deployment threshold they behave very differently. At train_p98, the VAE produces 176 false positives versus 1,916 for Isolation Forest (an **11 $\times$ reduction**), while missing 259 versus only 12 failure windows. The VAE’s missed detections cluster near the edges of failure intervals where the anomaly signature is partial, which is consistent with the last-10% window labeling rule.

We further observe that the val_f1 diagnostic result (F1 = 0.853) is close to train_p98 (F1 = 0.883), confirming that the 98th training-score percentile is a well-calibrated unsupervised threshold for this dataset.

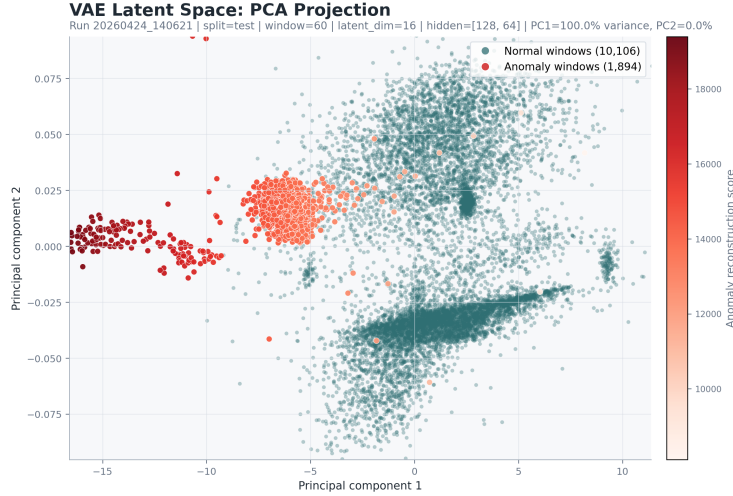


Figure 2: PCA projection of test-set latent means (VAE $[128, 64]$, $T = 60$). Normal windows (teal) and failure windows (orange–red, scored by reconstruction MSE) form largely distinct clusters in the latent space. The best-performing $[128, 64, 32]$ model’s posterior collapses near the prior ($\mu \approx \mathbf{0}$) under stronger KL regularization; reconstruction error is the operative anomaly signal in both models.

5 Conclusion

In this paper, we presented an unsupervised anomaly detection pipeline for industrial air compressors based on VAE reconstruction scoring. Under a train-only 98th-percentile threshold, our three-layer dense VAE substantially outperforms an Isolation Forest baseline on F1 (0.883 vs. 0.661) by reducing false positives eleven-fold, while both achieve identical ROC-AUC (0.996). Three practical lessons stand out: raw sensor magnitudes are essential anomaly signal and must not be scaled away; VAE training dynamics are sensitive to architecture in ways not fully controlled by the application seed; and equivalent ranking quality can coexist with dramatically different fixed-threshold behavior, so the choice of scoring method matters most precisely at the threshold a system is deployed at. Multi-seed evaluation and experiments on additional predictive maintenance benchmarks remain the most important directions for future work.

References

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A Window-Size Comparison

Table 5 reports test results across window lengths, all using the [128, 64] encoder. The 120-step window achieves ROC-AUC = 0.990 but low F1 (0.156) at train_p98, indicating well-ordered scores but a score-range mismatch with the fixed threshold. The 30- and 60-step runs collapse, consistent with the instability documented in the main text. We are cautious about attributing these differences to window length alone.

Table 5: Test results by window length (encoder [128, 64], train_p98).

Window	Prec.	Rec.	F1	ROC-AUC
30 steps	0.005	0.004	0.004	0.219
60 steps	0.174	0.060	0.089	0.091
120 steps	0.353	0.100	0.156	0.990

B Threshold Sensitivity

Table 6 shows the original primary VAE run ([128, 64], run 20260424_140621, trained before the grid study) across multiple threshold rules. We observe that train_p98 is the best unsupervised threshold and closely approximates the label-tuned val_f1 result (F1 = 0.821 vs. 0.792).

Table 6: Threshold sensitivity for the original [128, 64] VAE. Test split.

Threshold	Prec.	Rec.	F1	ROC-AUC
train_p95	0.213	0.993	0.351	0.988
train_p97	0.610	0.979	0.751	0.988
train_p98	0.716	0.963	0.821	0.988
train_p99	0.759	0.443	0.560	0.988
train_mean+3std	0.776	0.534	0.633	0.988
val_f1 (diag.)	0.793	0.791	0.792	0.988